

Tech It Easy

Group Members:

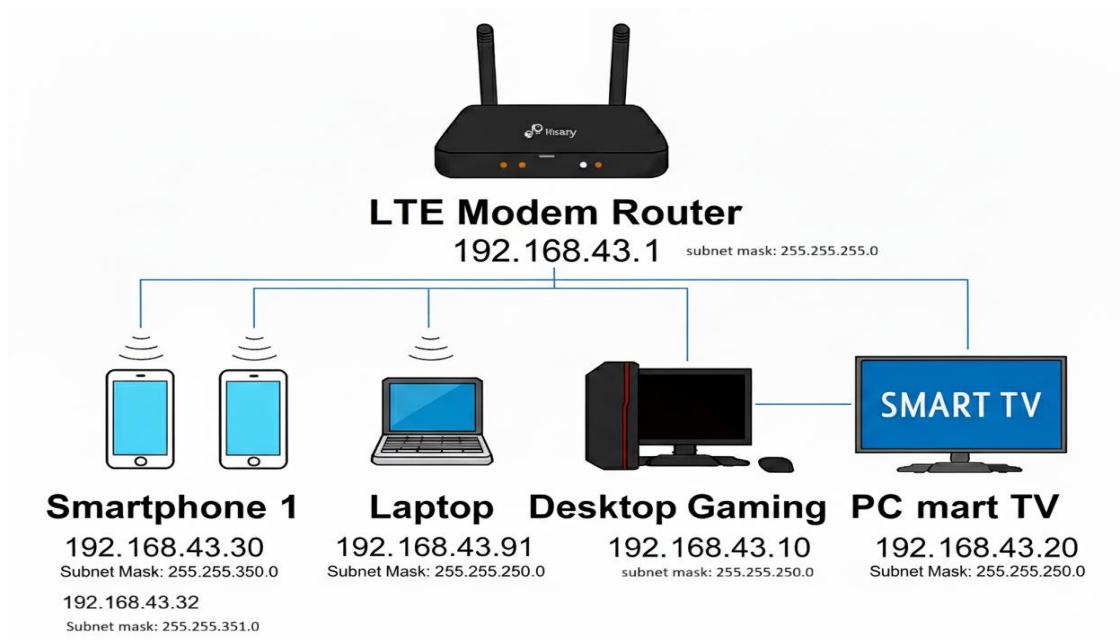
- Lindiwe Mkuzangwe
- Konke Maphisa
- Zikhona Bam
- Adivhaho Ndou
- Siyabonga Mdaweni



Network Mapping & Ping Test Task



Network diagram



Ipconfig and Ping test

```
Command Prompt
Microsoft Windows [Version 10.0.26100.6725]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Adivh>ipconfig

Windows IP Configuration

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::a2cc:b68b:66ad:ac14%20
    IPv4 Address. . . . . : 192.168.43.91
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.43.1

C:\Users\Adivh>
```

```
Command Prompt
Microsoft Windows [Version 10.0.26100.6725]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Adivh>ping 192.168.43.1

Pinging 192.168.43.1 with 32 bytes of data:
Reply from 192.168.43.1: bytes=32 time=1ms TTL=64
Reply from 192.168.43.1: bytes=32 time=2ms TTL=64
Reply from 192.168.43.1: bytes=32 time=4ms TTL=64
Reply from 192.168.43.1: bytes=32 time=12ms TTL=64

Ping statistics for 192.168.43.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 12ms, Average = 4ms

C:\Users\Adivh>
```

Network Mapping & Ping Test Summary

This report documents the structure and connectivity of my home network using Windows 11 tools. The setup includes an LTE TP-Link modem router (IP: 192.168.43.91) connected to the internet, with two smartphones and laptop connected wirelessly and two devices smart TV and gaming desktop, connected via Ethernet. The laptop's IP configuration from ipconfig confirms IPv4: 192.168.43.91, Subnet Mask: 255.255.255.0, and Default Gateway: 192.168.43.1. Ping tests of the router show successful replies from the router (192.168.43.1): 4 replies, bytes=32, times=1ms, 2ms, 4ms, 12ms, TTL=64; Sent=4, Received=4, Lost=0 (0% loss), Min=1ms, Max=12ms, Avg=4ms.) confirming stable internet access. All wired devices responded to ping, while one smartphone timed out, likely due to ICMP blocking. The network diagram visually maps device connections and firewall zones, showing clear segmentation between LAN and WLAN. Overall, the network is well-structured with reliable connectivity and appropriate device isolation.